Design Criteria Report — Single 10,000 US-gal 304 SS Water Storage Tank (Storage Only)

System: 10,000 US-gal covered 304 stainless steel storage tank for potable / process water (storage only).
Location (assumed): Charlotte, NC (preliminary geotechnical assumptions used).
Design status: Preliminary design criteria and planning-level cost estimate (2025 USD). Includes civil, containment, instrumentation, controls and minimal electrical. A 30% contingency applied to grand total.
Notes: All other design conditions remain the same as in the provided Salt Dilution System report and have been used where applicable (site slope, geotech assumption, codes/standards, containment requirement, etc.).

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1. Executive summary

Design for a storage-only system consisting of a single 10,000 US-gal covered vertical cylindrical tank fabricated from 304 stainless steel, elevated on a steel support platform (nominally 5 ft above final grade to match the template arrangement). Tank stores water at design temperature **60 °F** and atmospheric pressure. Instrumentation includes level transmitter, high/high alarm, temperature RTD, local PLC cabinet with HMI for monitoring and alarms. Secondary containment sized to **110%** of tank volume; stormwater detention sized to match the same 1-inch capture planning basis used in the template. Site, geotechnical, foundation, and permitting assumptions follow the template (Charlotte, NC preliminary values).

Estimated planning-level grand total (30% contingency included): **$~302,500 (2025 USD)** — itemized below. This is an order-of-magnitude planning estimate; final costs depend on vendor quotes, final platform complexity, piping choices, and local permitting. See Section 11 for itemized breakout.

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2. Design basis & explicit assumptions (imperial units)

Key design assumptions carried / adapted from the template:

- **Tank capacity:** 10,000 US-gal = 1,336.81 ft^3.
- **Fluid:** Water (no salt), design temperature = **60 °F**. Use water density at 60 °F ≈ **8.34 lb/gal** (SG ≈ 0.999 relative to the salt slurry case).
- **Operating pressure:** atmospheric (tank vented). No vacuum/pressure service assumed.
- **Tank orientation / aspect ratio:** keep **H:D = 2:1** as in the template to maintain similar footprint/height relationship (vertical cylindrical tank).
- **Elevation:** tank top / platform elevation **~5 ft above final grade** (matches earlier report).
- **Site / geotech:** preliminary allowable bearing = **3,000 psf** (Charlotte assumption). Final geotechnical investigation required.
- **Site slope & grading:** existing slope assumed **0.25%** for planning. Stormwater concept uses **1-inch capture** basis and 110% secondary containment as in the template.
- **Codes & standards:** ASME B31.3, AISC, ACI 318, NFPA (if applicable), NEC, OSHA, ASTM, NACE. Use local Charlotte/Mecklenburg stormwater & permitting rules.
- **Instrumentation / controls:** local PLC + HMI, level transmitter, high-high switch, temperature RTD, local alarm horn/strobe. User requested full instrumentation detail retained. 【5†source]}

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3. Tank — geometry, internals & utilities

3.1 Geometry (sizing calculation)

We adopt H:D = 2:1 and total volume = 10,000 US-gal.

- Convert gallons to cubic feet: 10,000 gal ÷ 7.48052 gal/ft^3 = **1,336.81 ft^3**. (See Appendix A for digit-by-digit arithmetic.)
- For a vertical cylinder with H = 2·D and volume V:
  ($V = \pi \cdot \frac{D^2}{4} \cdot H = \frac{\pi}{2} D^3$).
  Solve for D: ($D = \left(\dfrac{2V}{\pi}\right)^{1/3}$). With V = 1,336.81 ft^3 → D ≈ **9.48 ft**.
  Height H = 2·D ≈ **18.96 ft**. These match the geometry used in the template (for the same capacity and H:D).

3.2 Tank construction & internals

- **Material:** 304 stainless steel shell and dished head(s).
- **Cover:** Tight-fitting covered roof (weatherproof) with vent. Vent sized for atmospheric venting; include vent/overflow protection and screened inlet to avoid debris. (No flame arrestor required for water service.)
- **Manway:** 36-inch full access manway (top or shell as site layout dictates).
- **Drain:** 3-inch full-bore drain valve at tank bottom for emptying and sampling. Provide camlock or flange for portable transfer if needed.

- **No agitator or mixing internals** (storage-only). Include internal ladders/cages if required by code for >18 ft height and if personnel entry is expected.
- **Insulation / heat trace:** per template, heat trace and 1.5 in insulation are retained to maintain ~60 °F if freeze risk or process temperature control required. For unheated water storage in Charlotte, heat trace may be optional — included here as a planning allowance consistent with template practice.

3.3 Loads (loaded condition)

- **Fluid mass:** 10,000 gal × 8.34 lb/gal = **83,400 lb** (water at 60 °F). (See Appendix A for step-by-step arithmetic.)
- **Tank shell & internals (planning allowance): ~10,000 lb** (conservative allowance — includes shell, heads, fittings, manway, small ladders). Total loaded weight ≈ **93,400 lb**. (Template used a +10,000 lb allowance for similar tank; we mirror that assumption.)
- **Support reactions:** If supported on four pedestals, per-pedestal reaction ≈ 93,400 / 4 = **23,350 lb**.

3.4 No pumping package

- No fixed pump package or large suction/discharge piping is included in the system scope. Provide connection points (flanged outlets/inlets) for future temporary pumping or portable pump if the owner chooses to transfer water. Minimal piping allowances included in cost (Section 11).

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4. Support foundations, platform & civil summary

4.1 Platform & access

- **Platform elevation:** tank base/support at ~5 ft above final grade to match the template layout (access via stairs and landing). Provide stair, handrail, toe boards and OSHA-compliant grating and access platform. Platform and support steel sized for imposed loads and live loads per code.
- **Pedestals / pads:** Using preliminary allowable bearing **3,000 psf**: per-pedestal area required = reaction(23,350 lb) ÷ 3,000 psf = **7.783 ft²**. Use conservative **3 ft × 3 ft** pad (9 ft²) with 2 ft thickness reinforced concrete (same sizing convention as template). Foundation design to follow ACI 318; final sizing to be set after geotechnical report.
- **Platform framing & grating:** galvanized structural steel; connections and anchor bolts for concrete pads. Design per AISC and local codes.

4.2 Footprint & working area

- Minimal working footprint: equipment + platform + access = approximately **30 ft × 20 ft = 600 ft²** immediate footprint (tank, platform, access). Approximate project working area including containment, access lanes and temporary construction staging ≈ **6,000 ft²** to match typical minimal site requirements noted in the template (used for pavement, grading and stormwater planning).

4.3 Earthwork & excavation (planning)

- Excavation for four pads, trenching for small piping, and small platform footing. Planning earthwork volume (rough): **~150–600 cy** depending on approach and import/export for leveling. Use the template earthwork approach and refine after topographic survey.

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5. Containment, stormwater & grading

5.1 Secondary containment

- **Containment capacity requirement:** 110% of tank volume (template requirement) → 10,000 gal × 110% = **11,000 gal** = **1,466 ft³** ≈ **54.3 cy**. Provide coated concrete bermed containment or HDPE-lined containment basin sized to hold 11,000 gal with sump and recovery pump access. Materials chosen to resist corrosion; for water 304 SS compatibility is fine but coated concrete and HDPE liner recommended per template.

5.2 Stormwater (planning)

- Use the same **1-inch capture** basis for impervious area as template. For a working/impervious area estimate of ~4,800 ft², one-inch capture volume = 4,800 ft² × (1 in/12 ft) = **400 ft³** (~14.8 cy). Provide a small detention cell / infiltration basin / infiltration media sized accordingly and consistent with local Charlotte rules. Final dimensioning per stormwater calculations and local regs.

5.3 Grading & drainage

- Grade site for positive drainage (2% away from platform). Provide grassed swale / perimeter drain to route stormwater. Erosion control during construction per local standards. Template details apply. 【5†source}

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6. Instrumentation & controls

User requested similar instrumentation detail as in the template (but scaled down). Proposed list:

- **Tank level transmitter (4–20 mA) —** continuous level monitoring.
- **High level switch (discrete)** — high level alarm / shutoff interlock.
- **High-high level (discrete)** — safety alarm with local horn/strobe.
- **Temperature RTD** — monitoring tank temperature. (For water service and trace control.)
- **Local PLC cabinet with HMI (single-line overview)** — collects level/temperature/alarm signals and provides local alarm logging. Minimal I/O set (4 analog, 4 digital). Integrate with site BAS if required.
- **Field junction boxes, cable trays, grounding & lightning protection per NEC.**
- **Optional:** overflow switch, leak detection in containment sump.

Controls architecture: small PLC (compact unit), local HMI at platform or nearby kiosk, alarm relays, datalogging. Provide hardwired E-stop circuits for safe maintenance of platform. This mirrors the structure used in the template but simplified (no high-power VFDs).

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7. Electrical & small power summary

- **Motors:** none for major process (no pump or agitator). Small power only for instrumentation, heat trace circuits and containment sump pump (if owner chooses to install recovery pump). Include allowance for one small 3–5 HP sump pump if owner wants active spill recovery (optional).
- **Panel & distribution:** small local control panel (NEMA 3R or NEMA 4X depending on location), single-phase or three-phase tie-in depending on site preference. Typical sizing: **30–60 A** service if only instrumentation and trace; increase if sump pump included. Transformer sizing small — typical 10–25 kVA planning allowance.
- **Heat tracing:** self-regulating heat trace circuits and thermostat/overtemp protection; circuit loads sized per run lengths (planning allowance included).
- **Arc flash & coordination:** minor but still recommended to comply with NEC & NFPA 70E for any live panels. Template recommended a formal arc-flash study for larger motor loads — not required here but maintain good practice.

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8. Materials of construction, safety & access

- **Tank:** 304 stainless steel (shell & heads).
- **Piping for connections:** 304 SS for small diameter (inlet/discharge); major 316 SS not required for water. Use appropriate bolting and gaskets per ASTM/ASME standards.
- **Platform & structural steel:** galvanized steel per AISC, with stairways and handrails to OSHA.
- **Containment liner:** HDPE or coated concrete recommended (template used HDPE due to corrosive slurry — for water, epoxy coating is adequate but HDPE still acceptable).
- **Safety:** confined space signage, LOTO procedures, emergency access ladder, platform guards and toe boards, spill signage and response kit per template approach.

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9. Open items (required to finalize design)

These are gating items that must be resolved before proceeding to detailed design:

1. **Confirm fill/discharge strategy & desired turnover frequency** (owner to confirm typical fill/empty cycle, any bulk transfer requirements, and if a permanent sump pump is desired). This will affect piping, valves and any small pump selection.
2. **Confirm containment recovery strategy** — passive (siphon / vacuum truck) vs active (permanent recovery pump). A permanent recovery pump adds electrical and mechanical scope/cost.

3. **Final site coordinates & detailed topo** (for precise cut/fill, drainage and stormwater). Survey should be authorized immediately.
4. **Geotechnical report** (2 borings recommended). This will refine foundation design and allowable bearing. Template strongly recommended immediate geotech — we repeat that recommendation.
5. **Confirm local permitting & stormwater rules** (Charlotte/Mecklenburg) to finalize detention and pavement design.
6. **Decide insulation/heat trace necessity** (if owner elects to omit heat trace, remove allowance and reduce cost).
7. **Confirm electrical tie-in point & available voltage** (single-phase vs three-phase, transformer availability).

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10. Deliverables & schedule summary

Deliverables (planning):

- This Design Criteria Report (this document).
- Basic P&ID for tank connections and instrumentation.
- Tank GA drawing and support layout.
- Civil sketch for containment and grading.
- Equipment data sheet templates for tank vendor.
- Preliminary bill of materials (BOM) and procurement list.

Indicative schedule (planning):

- Kickoff & data confirmation (including authorizing topo & geotech): **2 weeks**.
- Vendor selection and tank procurement documentation: **4–8 weeks** (dependent on vendor lead times).
- Civil/site works (foundations, containment): **4–6 weeks**.
- Mechanical installation and commissioning: **2–4 weeks**.
- Total to ready for service (from data confirmation): **~10–18 weeks**, vendor lead times permitting.

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11. Preliminary cost estimate (itemized, 2025 USD, planning-level)

All numbers are planning-level and approximate. A 30% contingency is applied to the grand total.

11.1 Itemized pre-contingency estimates (planning)

- **10,000 gal 304 SS tank (fabrication, coatings, fittings, basic internals)**: **$60,000**
(Template had $70k for tank + agitator; agitator removed so reduced.)
- **Platform & structural steel (supports, stairs, handrail, grating, galvanize)**: **$40,000**
(Template used $60k for larger structure; scaled down for single tank.)
- **Concrete foundations & pedestals (4 pads + small slab work)**: **$12,000**
- **Small process piping & fittings (3" drain, flanged connections, valves, short run)**: **$15,000**

- **Instrumentation & controls (PLC, HMI, level TX, high/high switch, temp RTD, junction boxes, wiring)**: **$20,000**
  (Template used $75k for a larger system; scaled to tank-only.)
- **Electrical (panel, transformer allowance, heat trace circuits, cable tray)**: **$12,000**
- **Containment (110% lined containment basin with berms, sump)**: **$35,000**
  (Template sized 110% containment and recommended HDPE liner; this cost reflects small containment structure.)
- **Civil (survey, geotech, grading, minimal SWM allowance)**: **$26,000**
  - Survey & geotech (planning): ~$12,000 (template value mirrored).
  - Grading, minimal pavement & site prep: remainder.
- **Mechanical installation & field labor (installation, anchor bolts, sealing)**: **$25,000**
- **Indirects (engineering, permitting, commissioning allowance)**: **$15,000**

Subtotal (pre-contingency):
Sum = 60,000 + 40,000 + 12,000 + 15,000 + 20,000 + 12,000 + 35,000 + 26,000 + 25,000 + 15,000 = **$260,000**

11.2 Contingency & grand total

- **30% contingency:** 0.30 × $260,000 = **$78,000**
- **Grand total (with 30% contingency): $338,000**

11.3 Alternative (cost-reduced) scenario

If owner chooses to *omit* heat trace, use epoxy-coated containment rather than HDPE liner, and accept minimal instrumentation (level + local float alarm only), estimated pre-contingency could fall to **~$205k** and grand total with 30% contingency to **~$266.5k**. Final vendor quotes required for accuracy. The template used similar options to reduce cost (e.g., removing optional fire pump) — same approach applies.

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12. Codes, standards & references

Same as the template: ASME B31.3, AISC, ACI 318, AWS, ANSI/ASME flange specs, NEC (NFPA 70), OSHA 1910, ASTM and NACE material standards, and local Charlotte/Mecklenburg stormwater and permitting requirements. The template lists these standards and we follow them here.

Appendix A — Key calculations (digit-by-digit)

A.1 Volume conversions

- 1 US gallon = 0.133680556 ft^3 (or 1 ft^3 = 7.48052 US gal).
- 10,000 US-gal → ft^3:
  10,000 ÷ 7.48052 = 1,336.8101 ft^3 → round to **1,336.81 ft^3**. (Matches template conversion.)

A.2 Tank geometry (H:D = 2:1)

- $V = (\pi/2) \cdot D^3 \rightarrow D^3 = 2V/\pi$.
- Compute: 2 × 1,336.81 = 2,673.62.
- 2,673.62 ÷ 3.14159265 = 851.0086.
- Cube root(851.0086) = 9.480 ft (approx). → **D ≈ 9.48 ft**.
- H = 2·D = 18.96 ft → **H ≈ 18.96 ft**. (Same values used in the template for the same capacity / aspect ratio.)

A.3 Fluid weight

- Water density at 60 °F ≈ **8.34 lb/gal**.
- Mass = 10,000 gal × 8.34 lb/gal = **83,400 lb**.

A.4 Total loaded weight (planning allowance)

- Tank & internals allowance = **10,000 lb** (conservative).
- Total = 83,400 + 10,000 = **93,400 lb**.

A.5 Support reaction (4 pedestals)

- Per-pedestal reaction = 93,400 ÷ 4 = **23,350 lb**.
- Required pad area (prelim) at 3,000 psf allowable = 23,350 ÷ 3,000 = 7.783 ft^2 → choose 3 ft × 3 ft pad = 9 ft^2.

Appendix B — Preliminary instrument list (selected)

- 1 × Tank continuous level transmitter (4–20 mA)
- 1 × High level switch (discrete alarm)
- 1 × High-high level switch (safety alarm with horn/strobe)
- 1 × Temperature RTD (4–20 mA)
- 1 × Local PLC (compact) + HMI (local)
- 1 × Containment sump level